

# Le Huu Huy

 [github.com/lhhuy02012005](https://github.com/lhhuy02012005)  [Portfolio](#)  [hhuy.2005@gmail.com](mailto:hhuy.2005@gmail.com)

## EDUCATION

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| <b>Sai Gon University</b><br><i>B.S. Computer Science</i> | Step.2023 - June 2027<br><i>Current GPA: 3.16/4.0</i> |
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**Languages:** Java, PHP, Python, JavaScript/TypeScript, C/C++, C#, HTML/CSS  
**Frameworks & Libraries:** Spring Boot, Spring Data JPA, Hibernate, MapStruct, Laravel, React 19, FastAPI, LangChain  
**Security & IAM:** Spring Security, FusionAuth, OAuth2 (Google Login), JWT  
**Databases & Caching:** PostgreSQL, MySQL, Redis, ChromaDB, NoSQL  
**Architecture & Tools:** Docker, Git/GitHub, Unix Shell, Maven, Postman, WebSocket, Cloudinary

## PROJECTS

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| <b>ShopZues — Scalable E-commerce Platform</b>   <i>Spring boot, React, Firebase, MySQL, Redis</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Nov 2023 |
| <ul style="list-style-type: none"><li>Implemented a scalable Spring boot backend with JWT auth, RBAC (role/permission), and Redis (cache + queue + OTP).</li><li>Integrated Firebase Realtime DB for live order status (State Pattern) &amp; notifications, and FushionAuth for OTP email verification.</li><li>Developed React frontend + Flutter mobile app that connect to backend APIs and cloud DB (Firebase + MySQL).</li><li>Used Docker + docker-compose for consistent dev/test environments and easy deployment.</li><li>Learned end-to-end full-stack architecture: auth flows, real-time sync, payment flow.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |
| <b>RAGAI (NexusRAG) — Hybrid Knowledge Base &amp; Agentic Chat</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2026     |
| <ul style="list-style-type: none"><li><b>Tech Stack:</b> <i>FastAPI, React 19, TypeScript, PostgreSQL, ChromaDB, LightRAG, Ollama, Sentence-Transformers, Docker</i></li><li>Designed an end-to-end local-first RAG architecture: document upload → Docling parsing → chunking/dedup → embedding → vector + knowledge-graph indexing → hybrid retrieval → grounded generation.</li><li>Implemented a hybrid retrieval pipeline combining ChromaDB vector search and LightRAG graph context, then improved precision with cross-encoder reranking (<b>BAAI/bge-reranker-v2-m3</b>).</li><li>Built agentic chat with tool-calling (<b>search.documents</b>) and SSE streaming, delivering real-time events for thinking, tokens, sources, images, and final answer.</li><li>Developed citation-first response format with source traceability (document/page/heading path), enabling explainable answers and quick navigation back to evidence.</li><li>Added visual document intelligence: extracted images/tables from PDFs, generated captions, and injected media semantics into chunk embeddings for better retrieval quality.</li><li>Engineered robust document lifecycle handling (PARSING/INDEXING/INDEXED/FAILED), timeout recovery, and async background processing for long-running ingestion tasks.</li><li>Implemented workspace isolation across PostgreSQL metadata, vector collections, and knowledge-graph storage, supporting multi-tenant knowledge bases safely.</li><li>Containerized infrastructure with Docker Compose (PostgreSQL + ChromaDB) and tuned runtime configs (chunk size, prefetch, rerank top-k, KG extraction limits) for both medium and high-spec machines.</li></ul> |          |

### Sai Gon University

*Software Engineering Student*

Ho Chi Minh City, Viet Nam

*Sept. 2023 – Present*

- Developed high-performance web systems using **Laravel** and **Spring Boot**, focusing on Service-Repository patterns and clean architecture.
- Architected secure authentication systems using **JWT**, **OAuth 2.0**, and **Spring Security**, managing stateless sessions and role-based access control (RBAC).
- Optimized application performance by implementing **Redis** for distributed caching and asynchronous background tasks (queues), reducing database load during high-traffic simulations.
- Integrated third-party ecosystems including **VNPAY** for payment processing, **GHN** for logistics automation, and **Firebase** for real-time data synchronization.
- Containerized applications with **Docker** and orchestrated development environments using Docker Compose to ensure cross-platform consistency.
- Collaborated on version control via **Git/GitHub**, managing complex branching workflows and maintaining codebase integrity through code reviews.

## HOBBIES

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**Music & Arts:** **Playing badminton** , **Gym** , **Swimming**.

**Tech Exploration:** Researching **Generative AI (RAG)** and local LLM deployment; building custom mechanical keyboards.